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MILITARY SPECIFICATION

RADAR SET AN/APN-59B; BENCH, POST INSTALLATION, PREFLIGHT, AND FLIGHT TESTS OF

1. SCOPE

1.1 This specification covers methods and procedures for accomplishing the necessary performance and operation tests on the AN/APN-59B radar set, before and after installation in the aircraft, to determine that the equipment is in proper operating condition.

1.2 List of Articles for Test.- The articles to be tested according to procedures in this specification are as listed below. Wherever the type specification for a particular aircraft provides information not in agreement with this list, the list in the aircraft type specification shall govern. Any requirements herein which are applicable only to articles of equipment not covered in the list in the aircraft type specification shall be disregarded.

Receiver-Transmitter RT-289B/APN-59
Indicator, Azimuth and Range IP-239A/APN-59
Indicator, Azimuth and Range IP-268/APN-59
Power Supply PP-1073A/APN-59
Antenna AS-653A/APN-59 or Antenna AS-652A/APN-59
Amplifier, Electronic Control AM-853A/APN-59
Generator, Stabilization Data CN-221A/APN-59
Synchronizer, Electrical SN-198/APN-59
Control, Synchronizer C-2302/APN-59
Control, Radar Set C-1242/APN-59
Fan, Vaneaxial HD-150/APN-59

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

DRAWINGS

U.S. Air Force

54H2805	Installation Information - Cable Diagram; Radar Set AN/APN-59
58J4995	Cabling Diagram - AN/APN-59 Radar Set With Two IP-268/APN-59 Indicators

FSC 5841
FSC 6625

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. BENCH TESTS

3.1 The bench test procedure covers methods of checking the units of the AN/APN-59B radar set and the complete equipment on a test bench, at an initial installation or assembly point, prior to installation in an aircraft.

3.1.1 Power Requirements.- Unless otherwise specified, the power required for the bench test procedures specified herein shall be as follows:

a. AC Power Sources:

(1) 110V to 120V, 380 to 420 CPS, 2.5 amperes, single phase
(2) 110V to 120V, 380 to 1000 CPS, 10 amperes, single phase

b. DC Power Source: 25V to 29V, 10 amperes

3.1.1.1 If desired, the following AC power source may be used in place of those specified under 3.1.1a.: 110V to 120V, 380 to 420 CPS, 12.5 amperes, single phase.

3.1.2 Bench Layout and Facilities.- A bench shall be provided to accommodate the complete radar set and the test equipment without crowding. A minimum bench area of 40 square feet is desirable. Close proximity to a steel wall shall be avoided to afford protection of the magnetron.

3.1.2.1 If possible, the antenna shall be so located near a window that actual radar targets may be observed on the indicators.

3.1.2.2 Mounting fixtures for each radar set unit shall be provided for proper cooling and ease of maintenance.

3.1.2.2.1 Both external blowers of the receiver-transmitter shall be so mounted that each is within 18 inches of its respective endbell and the airstream will be within 30° of the receiver-transmitter centerline.

3.1.2.2.2 The stabilization data generator shall be mounted on the antenna.

3.1.2.3 The interconnecting cables used on the bench test setup and the lengths thereof shall be those used in the aircraft installation insofar as is practicable. Bench cable shall be in accordance with Drawing 54H2805. Bench cable for radar sets using the SN-198/APN-59 synchronizer shall be in accordance with Drawing 58J4995.

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3.2 Test Equipment Required.- Test equipment required for the bench tests shall be as follows, or the approved equivalent:

Description	Nomenclature	Applicable To
Oscilloscope	AN/USM-24	33A1-13-9-1
	TS-34AP	33A1-13-26-1
	Lavoie LS-239C	-----
FM Signal Generator	TS-147/UP	33A1-3-1-6
	TS-35A/AP	33A1-3-46-2
	TS-147B/UP	33A1-3-54-1
	AN/UPM-10A	33A1-3-10-11
Spectrum Analyzer	TS-148/UP	-----
Dummy Load	DA-22/U	-----
	TS-108/AP	33AA7-17-2
	Sperry 246	-----
Crystal Rectifier Test Set	TS-268/U	-----
Multimeter	AN/PSM-6	33A1-12-2-1
	TS-352/U	33A1-12-49-21
	IS-189	-----
	Simpson 260	-----
Audio Oscillator	TS-382B/U	33A1-8-81-1
	HP Model 201B	-----
Tube Tester	TV-2/U	-----
	TV-7/U	33AA21-5-1
	Hickok Model 539	-----
Tube Tester	I-177A	33AA21-3-1
	I-177B	33AA21-3-1
Tube Adapter Kit	MX-949/U	33AA21-1-104
Pressurizing Kit	MK-20A/AP	-----
Pressure Adapter	Airtron Model 16806A	-----
Electronic Multimeter	TS-505/U	-----
	TS-375/U	-----

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3.3 Test Procedures

3.3.1 Visual Inspection.- After the units have been unpacked, each unit shall be inspected externally for signs of shipping damage. This visual inspection shall include a check of all external parts such as controls, connectors, mountings, covers, and castings to assure that no shipping damage has occurred.

3.3.1.1 Dust Covers.- Dust covers shall not be removed unless visual inspection of the case indicates that internal damage may have occurred due to shipment or a tag on the unit indicates that the covers must be removed to remove desiccant bags, shipping clamps, or similar items prior to operation of the equipment.

3.3.1.1.1 The gyro cover shall be removed only in a dry, dust-free area.

3.3.2 Preliminary Procedures

a. Remove the dust covers necessary to perform the bench tests.

b. Connect bench test cabling to all units and connect waveguide from receiver-transmitter to antenna.

c. For optimum performance, the radar set shall not be operated on WARN function, 100/20 or 240/30-mile SEARCH function, or with the range delay switch in the ON (up) position during the first 10 minutes of radar operation.

3.3.3 Input Power.- After the time delay relays have closed and with the function switch in SEARCH position, the scan switch in PPI position, the range (NM) switch in "50-mile" range, and the stab switch in STAB (up) position, the input currents shall not exceed the following values:

a. 115V, variable frequency, 10.5 ±2 amperes

b. 115V, fixed frequency, 1.2 ±0.5 ampere

c. 26.5V DC, 4.9 ±1 ampere

3.3.3.1 A single fixed frequency source with a capacity of 2 KVA minimum may be used for the inputs specified in 3.3.3 a. and b.

3.3.4 Blowers.- The magnetron cooling blower shall force air out of the magnetron compartment of the receiver-transmitter unit. (The magnetron tube will be damaged if this blower is not operating.) The two external blowers shall blow onto the two endbells of the receiver-transmitter unit.

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3.3.4.1 The IP-239A/APN-59 indicator blower and Power Supply PP-1073A/APN-59 blower shall force air out of these units and there shall be no obstruction to this flow of air.

3.3.5 Edge Lighting.- Edge lighting on the following units shall operate:

a. C-1242/APN-59 radar set, control

b. IP-239A/APN-59 azimuth and range indicator

c. IP-268/APN-59 azimuth and range indicator

3.3.6 Receiver-Transmitter RT-289B/APN-59

3.3.6.1 Low Voltage Power Supply Output

3.3.6.1.1 With the function switch in SEARCH position, the range (NM) switch in the "50-mile" position, and the stab switch in the STAB (up) position, the low voltage power supply voltages as measured at the test jacks of the receiver-transmitter shall be as follows, when the variable frequency AC line voltage is 110V and when it is 120V:

a. Regulated 300V ±10V DC
b. Regulated 250V ±10V DC
c. Regulated 150V ±5V DC
d. Regulated to 105V ±5V DC
e. Regulated to 255V, +15V or -8V DC

3.3.6.1.2 With the same operating conditions as specified in 3.3.6.1.1, the unregulated 300V DC supply shall be 300V DC +15V to -45V at 115V AC line.

3.3.6.2 Pulse Repetition Frequency (PRF).- The PRF for each range and function shall be as follows:

a. Search, range delay switch in OFF (down) position:

(1) 3 to 30 mile, 2000 PPS ±10 percent
(2) 50 mile, 1025 PPS ±10 percent
(3) 100 and 240 mile, 180 PPS ±10 percent

b. Beacon, range delay switch in OFF (down) position:

(1) 3 to 30, 50 and 100 mile, 350 PPS ±10 percent
(2) 240 mile, 180 PPS ±10 percent

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c. Warning, range delay switch in OFF (down) position:

All ranges, 180 PPS ±10 percent

d. With the range delay switch in the ON (up) position, the PRF shall be 180 PPS ±10 percent for all positions, except OFF, of the range and function switches.

3.3.6.3 Magnetron Frequency and Spectrum

a. Frequency: 9375 ±40 MCS

b. The frequency spectrum width between 1/2 power points shall not exceed the following values:

Range	Search
3 to 30 mile	7.0 MCS
50 mile	3.5 MCS

c. Any side lobe outside the limits shown above shall be at least 6 DB down from the maximum spectrum amplitude as measured on the 3 to 30-mile range and with the receiver-transmitter RF output terminated with a dummy load.

3.3.6.4 Power Output

a. The RF output pulse shall have the following characteristics:

System Condition	Pulse Width 90 Percent Pt.
Search 3 to 30 mile	0.35 microsecond ±20 percent
Search 50 mile	0.80 microsecond ±10 percent
Search 100 mile	4.5 microsecond ±10 percent
Beacon 50 mile	2.35 microsecond ±0.15 microsecond

b. The average power output measured under the 100-mile search condition shall be 45 DBM minimum as measured with a TS-147/UP generator. For this test the receiver-transmitter shall be terminated with a dummy load.

3.3.6.5 Auxiliary Trigger Output.- With the system operating on 3 to 30-mile search range, the trigger voltage measured across a 500-ohm resistor connected to J-104 shall have a minimum peak amplitude of 50V and a minimum duration of 0.35 microsecond measured at 50 percent of the peak amplitude.

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3.3.6.6 Sensitivity.- With a test signal injected into J-121 of the receiver-transmitter, the system sensitivity shall be as follows (minimum discernible signal):

a. On 30-mile search: -99 DBM minimum
b. On 100-mile search: -104 DBM minimum
c. On 50-mile beacon: -94 DBM minimum

3.3.6.7 Crystal Protection Shutter Operation.- When the circuit is opened at pin C of J-502, the system sensitivity shall be decreased at least 25 DB.

3.3.6.8 Gain Control Range.- Limited noise shall appear on the indicators with clockwise rotation of the gain control on the radar set control.

3.3.6.9 Receiver Bandwidth.- When the range (NM) switch is changed from the "50-mile" position to the "100-mile" position and with the function switch in SEARCH, the bandwidth shall decrease.

3.3.6.10 Fast Time Constant (FTC).- Apply an RF signal to the system, just below limitation level as observed at J-607 when the FTC switch is in the OFF (down) position. When the FTC switch is thrown from the OFF to the ON (up) position, the signal observed at J-607 shall be differentiated.

3.3.6.11 Sensitivity Time Control (STC).- Place the function switch in the SEARCH position and the range (NM) switch in the "100-mile" position.

3.3.6.11.1 With the STC range control in the fully CLOCKWISE position, rotation of the STC amplitude control in a clockwise direction shall cause the noise on the cathode-ray tube sweep to decrease in intensity to 40 miles or beyond.

3.3.6.11.2 Starting with the STC range and amplitude controls set in the fully CLOCKWISE position, rotation of the range control in the counterclockwise direction shall gradually reduce the range over which the reduced noise is observed. When the range control is in the fully COUNTERCLOCKWISE position, no STC effect shall be observed.

3.3.6.12 Instantaneous Automatic Gain Control (IAGC).- With the gain control so set that noise is approximately 30 percent of limiting level, and with the system operating on a 3 to 30-mile search range and IAGC switch in the OFF (down) position, apply an RF signal whose duration falls between the limits of 0.5 to 3.0 microseconds. The signal strength shall be increased until the video pulse observed at J-607 limits. The signal input shall then be increased by approximately 35 DB. When the IAGC switch is thrown to "On", the pulse observed at J-209 shall be reduced by at least 3 DB.

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3.3.6.13 Radar Automatic Frequency Control (AFC).- With the function switch in SEARCH position, the range (NM) switch in the "100-mile" position, and the test meter switch in the "AFC-1" and "AFC-2" position in turn, the meter readings shall each be 0.7 ±0.2 MA, and the difference between the two readings shall not exceed 0.2 MA. In the event that malfunctioning of the AFC circuit is encountered, refer to the applicable AN/APN-59B technical order handbook for adjustment procedures.

3.3.6.14 Beacon AFC.- With the function switch in BEACON position, the range (NM) switch in any position, and the test meter switch to the "MIX-1" and "MIX-2" in turn, the meter readings shall each be 0.7 ±0.2 MA. The difference between the two readings shall not exceed 0.2 MA. In the event that malfunctioning of the Beacon AFC circuit is encountered, refer to the applicable AN/APN-59B technical order handbook for adjustment procedures.

3.3.6.15 Test Meter

3.3.6.15.1 With the test meter switch in the LINE position, and with the variable frequency input line voltage adjusted to 115V, the test meter shall indicate 0.6 ±0.05.

3.3.6.15.2 With the test meter switch in MAG position, the meter shall indicate magnetron current. At any given range, the meter reading shall be steady. A slight instability for a period of up to three seconds following a change in setting of the range (NM) or function switch as well as an infrequent jump in the indication during operation on any range is permissible.

3.3.7 Indicator, Azimuth and Range IP-239A/APN-59

3.3.7.1 Intensity Control (INT).- With the function switch at STDBY, maximum clockwise position of INT control shall produce a spot of low intensity. With the function switch at any position other than STDBY, maximum clockwise position of the INT control shall produce a bright trace and maximum counterclockwise position shall eliminate saturated signals.

3.3.7.2 Focus Control.- The focus control shall permit optimum focus at other than the extreme limits of its travel. The sweep shall be defocused at both extreme positions of the focus control.

3.3.7.3 Range Marker Lights.- The range marker lights shall indicate the position of the range (NM) switch.

3.3.7.3.1 The red light marked "TD" shall light and the 3-30/5 mile light shall light, when the range delay switch is in the ON position.

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3.3.7.4 Range Mark Intensity Control (RANGE MKS).- With the gain control on the radar set control so rotated that noise appears on the indicator, a clockwise rotation of the range mark intensity control shall increase the range marks to a clearly visible intensity.

3.3.7.5 Sweeps and Fixed Range Markers

3.3.7.5.1 Each range sweep shall start at the same point, within 1/16 inch of the cursor cross hairs, when the sweep is in focus. No "hole" or reverse sweep greater than 1/8 inch in diameter shall be present. For all ranges the sweep length shall not be less than two inches.

3.3.7.5.2 The appropriate number of range marks shall appear for each setting of the range (NM) switch, and shall be as follows:

Range	Number of Range Marks
3-30/1 minimum setting	2 or 3
3-30/1 maximum setting	25 minimum
3-30/5 minimum setting	none
3-30/5 maximum setting	5 or 6
50/10	4 or 5
100/20	5 or 6
240/30	8 or 9

In the event that malfunction is encountered, refer to the applicable technical order of the AN/APN-59B for adjustment procedures.

3.3.7.5.3 Range mark ripple shall not be in excess of three times the range mark width.

3.3.7.5.4 The relationship of the 1, 5, 10, 20, and 30-mile range marks shall be correct.

3.3.7.6 Range Delay Controls and Variable Range Marker

3.3.7.6.1 With the range delay switch turned to the OFF (down) position, the range delay control shall vary the delay range mark over a range starting at not more than 15 miles and ending at not less than 210 miles. The most counterclockwise position of the range delay control that produces the variable range mark shall be used. Clockwise rotation of the control shall increase the range of the mark.

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3.3.7.6.2 Ripple on the variable range mark shall not be in excess of three times the range mark width.

3.3.7.6.3 When the range delay switch is turned "on", a sweep adjustable to 25 miles or greater shall be produced, starting at the range indicated by the variable mark. This function shall be checked at approximately 20 miles on the 50-mile range and approximately 200 miles on the 240-mile range. With the range delay control set to less than 15 miles and the TD switch "on", the sweep may disappear.

3.3.7.7 Dial Dim Control and Heading Marker Intensity Control

3.3.7.7.1 Dial Dim Control.- The intensity of the azimuth ring on the front panel of the indicator shall be adjustable by the dial dim control.

3.3.7.7.2 Heading Marker Intensity Control.- With the HDG MKR control in a fully clockwise position, a dark or "black" heading marker shall be obtained. As the control is moved in a counterclockwise direction, the dark marker shall first disappear and shall be replaced by an increasingly bright or "white" marker as the control is moved to the fully counterclockwise position.

3.3.7.8 Operation With Doppler Drift Attachment AN/APA-52

3.3.7.8.1 With the fore-normal-aft switch on Doppler Drift Attachment AN/APA-52 in FORE position, antenna positioning shall be controllable with the drift angle dial on the AN/APA-52. Clockwise rotation of the drift angle control shall produce a clockwise rotation of the indicator trace.

3.3.7.8.2 With the fore-normal-aft switch in NORMAL position, an azimuth marker produced by the AN/APA-52 shall be observable on IP-239A/APN-59 indicator and its intensity shall be variable by means of the AZ mark intensity control located on the AN/APA-52.

3.3.7.9 Video Test Output.- The video signal (as observed at J-607) of the indicator, on the 50-mile range, shall have a limiting level between 20V and 35V as measured from the top of the pedestal.

3.3.8 Indicator, Azimuth and Range IP-268/APN-59 and Power Supply PP-1073A/APN-59.- The auxiliary indicator with its power supply shall be in accordance with the requirements of 3.3.7.1, 3.3.7.2, 3.3.7.3, 3.3.7.5 (3.3.7.5.1, 3.3.7.5.2, and 3.3.7.5.3), and 3.3.7.7.

3.3.8.1 Slave-Relative Switch

3.3.8.1.1 With this switch in first the "S" position and then "R" position and the antenna in fast scan, the indicator trace shall rotate in a clockwise direction.

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3.3.8.1.2 With this switch in the "S" position and the bearing switch in "REL" position, the heading marker shall appear at a position within ±2° of 0° on the IP-268/APN-59.

3.3.8.1.3 With this switch in the "S" position and the bearing switch in STAB position and with the HDG-SEL control set to position the marker at 90° on the IP-239A/APN-59, the heading marker shall appear within ±2° of 90° on the IP-268/APN-59.

3.3.8.1.4 With this switch in the "R" position, the heading marker shall appear within ±2° of 0° on the IP-268/APN-59 regardless of the position of the BEARING switch.

3.3.9 Antenna AS-653A/APN-59 or Antenna AS-652A/APN-59; Amplifier, Electronic Control AM-853A/APN-59, and Generator, Stabilization Data CN-221A/APN-59

3.3.9.1 Antenna Rotation and Speed

3.3.9.1.1 With the scan switch in "L" position, the antenna and the two indicator traces shall rotate in a counterclockwise direction at the slow rate (10 to 15 RPM).

3.3.9.1.2 With the scan switch in the "R" position, the antenna and the two indicator traces shall rotate in a clockwise direction at the slow rate (10 to 15 RPM).

3.3.9.1.3 With the scan switch in the OFF position, rotation of the antenna and the two indicator traces shall stop.

3.3.9.1.4 With scan switch in the PPI position and the function switch in SEARCH position, the antenna shall scan at the fast rate (45 ±7 RPM) when the range (NM) switch is in the 3-30/1, 3-30/5 and 50/10 positions. The antenna shall rotate at the slow rate for all other positions, except OFF, of the range and function switches.

3.3.9.1.5 With the scan switch in SECTOR position, the antenna shall sector scan an angle of 81° ±10°* centered about a "Dead Ahead" position. The sector as observed on the two indicators shall be centered within ±10° of 0°.

*Note: This is the sector scan angle as shipped from the factory and it may be changed to suit the requirements of a particular aircraft installation.

3.3.9.2 Antenna Tilt.- When the position of the tilt control is varied between its limits, the antenna reflectors shall follow. Positive reflector tilt shall occur for positive control settings.

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3.3.9.3 Antenna Beam Pattern Change.- When the PATT switch is in PENCIL position, the parabolic reflector shall be in operating position. With the tilt control in the maximum positive position, the cosecant-squared shaped reflector of the antenna shall rotate into the operating position when the PATT switch is thrown to the FAN position. The reflector tilt shall then return to the maximum positive position after the beam switching sequence has taken place.

3.3.9.4 Erection Voltage Cutoff

3.3.9.4.1 When the roll angle of the antenna is at any angle between 0° and 8°, erection voltage measured between JB2-3 and JB2-42 shall be 25V AC or greater.

3.3.9.4.2 When the roll angle of the antenna is at any angle exceeding 12°, the erection voltage measured between JB2-3 and JB2-42 shall be zero.

3.3.9.5 Stabilization.- Level the antenna base and level the stabilization generator on the tilt table in pitch and roll. Place antenna tilt on ZERO and STAB ON.

a. Pitch the stabilization generator to a 10° nose down position.

b. Rotate antenna reflectors to "Dead Ahead". Antenna feed horn shall point up 10° ±2° from original position.

c. Rotate antenna reflectors to 180° point. Antenna feed horn shall point down 10° ±2° from original position.

d. Return stabilization generator to level position.

e. Roll the stabilization generator 10°, right wing down.

f. Rotate antenna reflectors to 90°. Antenna feed horn shall point up 10° ±2° from original position.

g. Rotate antenna reflectors to 270°. Antenna feed horn shall point down 10° ±2° from original position.

3.3.9.6 Gimbal Caging.- With the antenna pitched approximately 6° and rolled approximately 12°, and the antenna in fast scan, and when the STAB switch is thrown from ON (up) position to OFF (down) position, the stabilization circuits shall become inoperative and the reflectors shall be brought to the caged position.

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3.3.9.7 Heading Marker and Azimuth Accuracy

3.3.9.7.1 With the heading reference marks on the antenna aligned, the antenna stopped, and the bearing switch in REL position and the S-R switch in "S" position, the cathode-ray tube traces shall indicate 0° ±2° on both indicators.

3.3.9.7.2 With the antenna scanning at the fast rate, the heading marker on each indicator shall occur at 0° ±2°.

3.3.9.7.3 With the antenna sector scanning, the heading marker on each indicator shall occur at 0° ±2°.

3.3.9.8 Heading Selector

3.3.9.8.1 With the antenna stopped, the bearing switch in STAB position, the S-R switch in "S" position, the HDG SEL control set to indicate 0°, and the compass simulator adjusted to 0°, the trace on each indicator shall indicate 0° ±2° when the heading reference marks on the antenna are aligned.

3.3.9.8.2 As the HDG SEL control is rotated in a clockwise direction, the two indicator traces shall rotate in a counterclockwise direction.

3.3.9.8.3 With the HDG SEL control set at 90° the trace on each indicator shall indicate 280° ±2°.

3.3.10 Synchronizer, Electrical SN-198/APN-59 and Control, Synchronizer C-2302/APN-59.- When used in conjunction with the IP-268/APN-59 indicator in lieu of the IP-239A/APN-59 indicator, the requirements for the SN-198/APN-59 synchronizer and the C-2302/APN-59 control shall be in accordance with 3.3.7.3.1, 3.3.7.4, 3.3.7.5, and 3.3.7.6.

3.3.10.1 Heading Marker Intensity Control.- With the HDG MKR control in a fully clockwise position, a dark or "black" heading marker shall be obtained. As the control is moved in a counterclockwise direction, the dark marker shall first disappear and then shall be replaced by an increasingly bright or "white" marker as the control is moved to the fully counterclockwise position.

3.3.11 Pressure Tests

3.3.11.1 System Pressure Test.- Pressurize system to 15 PSI gage. Limit waveguide lengths between the receiver-transmitter and antenna to 6 feet and insure that gaskets have been placed between interconnecting joints. After 5 minutes, with the antenna rotation stopped, gage reading shall not be less than 14 PSI gage.

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3.3.11.2 Receiver-Transmitter Pressure Test.- With the receiver-transmitter pressurized to 15 PSI gage, pressure shall not be less than 14 PSI gage after 50 minutes.

3.3.11.3 Antenna Pressure Test.- With the antenna pressurized to 15 PSI gage and antenna rotation stopped, pressure shall not be less than 2 PSI gage after 2 minutes.

4. POST INSTALLATION TESTS

4.1 The post installation test procedure covers requirements for checking Radar Set AN/APN-59B after it has been installed in the aircraft.

4.1.1 Power Requirements.- The radar set shall be operated within the following limits of line voltage and frequency:

a. AC Power:

(1) 115V ±5V, 380 to 1000 CPS
(2) 115V ±5V, 380 to 420 CPS

b. DC Power: 25V to 29V

4.1.1.1 A single AC power source, the frequency limits of which do not exceed 380 to 420 CPS, may be used in lieu of the two AC sources specified under 4.1.1a.

4.2 Test Equipment Required.- Test equipment required for the post installation tests shall be as follows, or the approved equivalent:

Description	Nomenclature
FM Signal Generator	TS-147/UP
Spectrum Analyzer	TS-148/UP

4.3 Test Procedures

4.3.1 Visual Inspection

a. Fuses.- Check all fuses on receiver-transmitter and Power Supply PP-1073A/APN-59. Spare fuses shall be provided.

b. Waveguide.- Waveguides shall show no evidence of damage. Screws holding flange-to-choke joints shall be tight.

c. Isolators, shock mounts and bonding straps.- Isolators, shock mounts and bonding straps shall be properly installed.

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d. Connectors.- All connectors shall be tight. Cable connections shall be correctly made, in particular the connections to J-2904 and J-2906 on PP-1073A/APN-59 must be correctly made to prevent damage to the equipment.

e. Radome.- The radome shall be free of damage, pits, soft spots, etc.

f. Generator, Stabilization Data Installation.- Generator, Stabilization Data CN-221A/APN-59 shall be installed so that the arrow on the gyroscope is pointed ahead.

4.3.2 The post installation tests shall include the following:

TESTS	APPLICABLE PARAGRAPH
Blowers	3.3.4
Edge Lighting	3.3.5
Low Voltage Power Supply Output	3.3.6.1
Magnetron Frequency and Spectrum	3.3.6.3a. and 3.3.6.3b.
Power Output	3.3.6.4a.
Sensitivity	3.3.6.6
Fast Time Constant (FTC)	3.3.6.10
Sensitivity Time Control (STC)	3.3.6.11
Radar AFC	3.3.6.13
Beacon AFC (Beacon local oscillator frequency need not be checked)	3.3.6.14
Test Meter	3.3.6.15
Intensity Control	3.3.7.1
Focus Control	3.3.7.2
Range Marker Lights	3.3.7.3
Range Mark Intensity Control	3.3.7.4
Sweeps and Fixed Range Markers	3.3.7.5

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TESTS	APPLICABLE PARAGRAPH
Range Delay Controls and Variable Range Marker	3.3.7.6
Dial Dim Control and Heading Marker Intensity Control	3.3.7.7
Operation With AN/APA-52	3.3.7.8
Indicator, Azimuth and Range IP-268/APN-59 and Power Supply PP-1073A/APN-59	3.3.8
Antenna Rotation and Speed	3.3.9.1
Antenna Tilt	3.3.9.2
Antenna Beam Pattern Change	3.3.9.3
Erection Voltage Cutoff (This test may be made by tilting the gyro case to the angles specified)	3.3.9.4
Synchronizer, Electrical SN-198/APN-59 and Control, Synchronizer C-2302/APN-59	3.3.10

4.3.2.1 Magnetron Spectrum.- Any frequency spectrum side lobes outside of the limits given in 3.3.6.3b. for the 3 to 30-mile range shall be at least 6 DB down from the maximum spectrum amplitude.

4.3.2.2 Power Output.- The average power output measured on the 100-mile search position shall be 45 DBM minimum, as measured with a TS-147/UP generator.

4.3.3 System Pressure Test.- The pressurization system shall maintain system pressure between 27 and 32 inches Hg.

4.3.4 Antenna Stabilization.- With the STAB switch ON (toggle up) and with the scan switch in the OFF position, antenna looking straight ahead, disconnect gyro from its mounting surface and move gyro so as to simulate a roll and pitch movement of the aircraft. The antenna reflectors shall respond in a direction opposite to this movement of the gyro case.

4.3.5 Azimuth Stabilization and Heading Selector Control

4.3.5.1 With bearing switch in STAB position, HDG-SEL control set to 0°, S-R switch on IP-268/APN-59 set to "S" position, and with the antenna operating in either slow or fast scan, the heading markers on both indicators shall indicate compass heading within ±2°.

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4.3.5.2 With a 90° clockwise rotation of the HDG-SEL control, both indicator traces shall rotate 90° ±2° in a counterclockwise direction.

4.3.6 Heading Marker and Azimuth Accuracy

4.3.6.1 With the antenna stopped in the "Dead Ahead" position, the bearing switch in REL position, and the S-R switch in "S" position, the cathode-ray tube traces shall indicate 0° ±2° on both indicators.

4.3.6.2 With the antenna scanning at the fast rate, the heading marker on each indicator shall occur at 0° ±2°.

4.3.6.3 With the antenna sector scanning, the heading marker on each indicator shall occur at 0° ±2°.

5. PREFLIGHT TEST

5.1 The preflight test procedure covers methods of checking the installed Radar Set AN/APN-59 prior to aircraft flights.

5.1.1 Power Requirements.- The radar shall be operated within the following limits of line voltage and frequency:

a. AC Power:

(1) 115V ±5V, 380 to 1000 CPS
(2) 115V ±5V, 380 to 420 CPS

b. DC Power: 25V to 29V

5.1.1.1 A single 115V ±5V AC power source, the frequency limits of which do not exceed 380 to 420 CPS, may be used in lieu of the two AC sources specified under 5.1.1a.

5.2 Test Equipment Required.- No test equipment required.

5.3 Test Procedures

5.3.1 Visual Inspection.- All units, cables, connectors, shock mounts, and viewing hoods that are accessible shall be visually inspected.

5.3.2 Pressure Test.- System pressure test shall be in accordance with 4.3.3.

5.3.3 Blower Operation.- Blower operation shall be in accordance with 3.3.4.

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5.3.4 Operation of the following controls shall be as specified:

	CONTROLS	APPLICABLE PARAGRAPH
a.	Intensity Control	3.3.7.1
b.	Focus Control	3.3.7.2
c.	Range Mark Intensity Control	3.3.7.4
d.	Range Delay Controls and Variable Range Marker	3.3.7.6
e.	Dial Dim Control	3.3.7.7.1
f.	Heading Marker Intensity Control	3.3.7.7.2 3.3.10.1
g.	Slave-Relative Switch	3.3.8.1
h.	Antenna Rotation and Speed	3.3.9.1
i.	Antenna Tilt	3.3.9.2
j.	Antenna Beam Pattern Change Switch	3.3.9.3
k.	Gain Control	3.3.6.8
l.	STC Control	3.3.6.11
m.	Test Meter (All positions)	3.3.6.13, 3.3.6.14, and 3.3.6.15

5.3.5 FTC Switch.- With the use of the FTC switch, the presentation shall be broken up into more individual and more readily distinguished targets. This test shall be made with the IAGC switch in the OFF position.

5.3.6 Range Marker Lights.- Range marker lights shall be in accordance with 3.3.7.3.

5.3.7 Sweeps and Fixed Range Markers.- Sweeps and fixed range markers shall be in accordance with 3.3.7.5.

5.3.8 Operation with AN/APA-52.- Operation with AN/APA-52 shall be in accordance with 3.3.7.8.

5.3.9 Heading Marker and Azimuth Accuracy.- Heading marker and azimuth accuracy shall be in accordance with 3.3.9.7.2 and 3.3.9.7.3.

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5.3.10 Azimuth Stabilization and Heading Selector Control.- Azimuth stabilization and heading selector control shall be in accordance with 4.3.5.

6. FLIGHT TESTS:

6.1 The flight test procedure covers requirements to operationally check Radar Set AN/APN-59B in flight.

6.1.1 Power Requirements.- The radar set shall be operated within the following limits of line voltage and frequency:

a. AC Power:

(1) 115V ±5V, 380 to 420 CPS
(2) 115V ±5V, 380 to 1000 CPS

b. DC Power: 25V to 29V

6.1.1.1 A single 115V ±5V AC power source, the frequency limits of which do not exceed 380 to 420 CPS, may be used in lieu of the two AC sources specified under 6.1.1a.

6.2 Test Equipment Required.- No test equipment required.

6.3 Test Procedures

6.3.1 Preliminary Adjustment.- For optimum performance, the radar set shall not be operated on WARN function, 100/20 or 240/30-mile SEARCH function, or with the range delay switch in ON (up) position during the first 10 minutes of radar operation.

6.3.2 Pressure Test.- The pressurization system shall be maintained between 27 and 32 inches Hg throughout the flight.

6.3.3 Warmup Period.- Magnetron current shall be indicated on the test meter within 5 minutes after the function switch is turned from OFF to SEARCH position.

6.3.4 Test Meter.- The test meter reading shall be as follows:

TEST METER SWITCH POSITION	READING
LINE	0.6 ±0.08
MAG.	Approximately 0.6*
MIX 1	0.6 ±0.3
MIX 2	0.6 ±0.3
AFC 1	0.6 ±0.3**
AFC 2	0.6 ±0.3**

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6.3.4.1 *Considerable variation of this nominal value may be expected from system to system and from range to range. At any given range the meter reading should be steady. A slight instability for a period of up to approximately three seconds following a change in setting of the range or function switch as well as an infrequent jump in the indication during operation on any range is permissible.

6.3.4.2 **These readings required for SEARCH function only.

6.3.5 Operating Controls.- All controls specified under 5.3.4 shall operate properly.

6.3.6 Range Marker Lights of IP-239A/APN-59 and IP-268/APN-59.- Range marker lights of IP-239A/APN-59 and IP-268/APN-59 indicators shall be in accordance with 3.3.7.3.

6.3.7 Sweeps and Fixed Range Markers.- Sweeps and fixed range markers shall be in accordance with 3.3.7.5.

6.3.8 Heading Marker and Azimuth Accuracy.- Heading marker and azimuth accuracy shall be in accordance with 3.3.9.7.2 and 3.3.9.7.3.

6.3.9 Azimuth Stabilization and Heading Selector Control.- Azimuth stabilization and heading selector control shall be in accordance with 4.3.5.

6.3.10 Fast Time Constant (FTC).- The FTC switch shall be in accordance with 5.3.5.

6.3.11 Sensitivity Time Control (STC).- With the range control in the 100 or 240-mile position, the use of the STC controls shall break up solid returns at ranges of 30 miles and less into more individual returns.

6.3.12 Instantaneous Automatic Gain Control (IAGC).- With the use of the IAGC switch an effect similar to that produced by the FTC shall be noted. This test shall be made with the FTC switch in the OFF position.

6.3.13 Operation with Doppler Drift Attachment AN/APA-52.- Operation with AN/APA-52 shall be in accordance with 3.3.7.8.

6.3.14 Radar Range, Fan Beam.- Ground mapping with a well defined presentation to a minimum range of 90 miles shall be obtained at altitudes in excess of 10,000 feet when using fan beam.

6.3.15 Radar Range, Pencil Beam.- Ground mapping with a well defined presentation to a minimum range of 110 miles shall be obtained at altitudes in excess of 10,000 feet when using pencil beam.

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6.3.15.1 Observe that the concentration of energy in the pencil beam will have the following effects:

a. Weak returns that were detectable at maximum range with fan beam will appear stronger.

b. More distant targets will be visible.

c. The area at short ranges is not covered by the beam and consequently the number of targets observable at short ranges will be considerably reduced.

6.3.16 Beacon Range.- At altitudes in excess of 10,000 feet, beacon replies shall be obtained at a minimum range of 100 miles with the radar operating in conjunction with Radio Set AN/CPN-6 (racon station) or equivalent.

6.3.16.1 It shall be possible to obtain the beacon station identification by use of the delayed sweep and 3 to 30 range controls.

6.3.16.2 When the radar is turned to beacon function, the beacon AFC circuits may require a warmup period of approximately 15 seconds before normal operation is obtained.

6.3.17 Warning Function.- With the function selector in WARN position, weather formations shall be detected whose intensity will vary depending upon range and precipitation content. (Omit in absence of weather.)

6.3.18 Antenna Stabilization.- At an altitude of 10,000 feet and PPI operation and with the target display limited to 40 miles by the tilt control, mapping shall not decrease to less than 30 miles in any sector (which is normally mapped) as the aircraft is maneuvered through a 6-minute, 360-degree turn with a bank angle between 20° and 25°. The aircraft shall maintain straight and level altitude for 3 to 5 minutes prior to this test.

7. NOTES

7.1 Intended Use.- Radar Set AN/APN-59B is a compact, light weight, high powered radar designed for high or low altitude navigation. Its functions include mapping, terrain clearance, warning for storm avoidance, and beacon.

7.2 Normal Operation.- The following paragraphs describe conditions of operation of the AN/APN-59B radar which are normal and shall not be cause for rejection.

7.2.1 Sweep Centering.- Adjustment of the focus control will affect sweep centering. The degree to which centering is affected by the focus control may vary from unit to unit. The sweep shall be centered when the cathode-ray tube is in focus.

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7.2.2 Heading Marker Visibility.- It is not necessary to obtain a continuous and uniformly intense heading marker through areas exhibiting both strong and little or no radar return. The heading marker may not be visible over areas of strong radar return or in areas of no radar return depending upon the setting of gain and intensity controls. With these controls adjusted for optimum radar return, less than optimum heading marker intensity may result.

7.2.3 Rings of No Return.- Rings of no return observed on the indicator when using fan beam are acceptable provided that targets can be presented in these areas by adjustment of controls available to the operator. On the 50-nautical mile range, targets may be lost in the outer range of the scope due to these adjustments.

7.2.4 Altitude Line.- A well defined altitude line should not be expected under all conditions of operation particularly at high altitudes. On the 50-nautical mile range, the radar set is acceptable with no targets inside of a slant range equal to the absolute altitude of the aircraft plus 8 nautical miles, at aircraft altitudes above 10,000 feet.

7.2.5 Pencil Line Spoking.- The indicator may exhibit a uniform pattern of fine and uniform radial lines through the presentation.

7.2.6 Variable Range Marker.- The variable range marker intensity will usually be less on long ranges than the intensity of the fixed markers. Variable range marks shall be clearly visible on long range beacon operation but may be obscured under some conditions by strong radar return on Search and Warning presentations.

7.2.7 Range Marker Distortion.- A slight notch may be seen on the range marks immediately following the heading marker. The range markers may also appear slightly distorted near areas of very strong signal return. That is, the marker will appear slightly further from the center of the presentation.

7.2.8 Sweep Instability.- When the AN/APN-59B radar is operated on the 3-30/1 mile range, there is a position of the 3-30 range control in which a slight sudden change of sweep length is obtained. That is, the sweep length versus control position has a slight discontinuity which actually occurs when the range is set somewhere between 4 and 7 miles. The above condition is indicated by a sudden displacement of the 1 mile range marks as the range knob is moved slightly in the region where the discontinuity exists. A setting can sometimes be obtained which produces double range marks. If the range control knob is turned slightly clockwise or counterclockwise from the position which produces the above condition, an entirely satisfactory presentation shall be obtained.

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7.2.8.1 Another condition of sweep instability can be produced on the 3-30 mile range on some units when the 3-30 range control is adjusted to the maximum clockwise position. This sweep instability will occur only on units in which this control will produce a range display in excess of 30 miles. The presentation should be satisfactory however, as long as the 3-30 mile range is kept below 30 miles.

7.2.9 When the radar set is turned from BEACON to RADAR function, the radar local oscillator may require a warmup period of up to one minute before normal AFC operation is obtained.

7.3 It is necessary that personnel making tests described herein be thoroughly familiar with the operating characteristics of Radar Set AN/APN-59. The following handbooks may be of interest in connection with this work.

T.O. 12P5-2	APN 59-1	Handbook Operating Instructions
T.O. 12P5-2	APN 59-2	Handbook Service Instructions
T.O. 12P5-2	APN 59-3	Handbook Overhaul Instructions
T.O. 12P5-2	APN 59-4	Illustrated Parts Breakdown

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